How accurate is LLM feedback on second language learners' errors?

LLMs provide **accurate surface feedback**, but struggle with overcorrection and discourse.

Is LLM feedback accurate for second language learners errors? N = 14



Yes 64% · Possibly 0% · Mixed 21% · No 14%

Large language models demonstrate strong precision for detecting and explaining local grammatical, lexical, and morphological errors, but their reliability decreases on higher-order discourse structures and low-resource languages (Craven & Fredrick, 2026; Jaganov et al., 2025; Pasha, 2026; Lin, 2024).

FIGURE 1 Scientific consensus on the accuracy of LLM feedback for second language errors

Effectiveness of LLMs on Language Learner Errors

Evidence Strength	Claim
Strong	LLMs achieve **high precision on surface-level errors**—such as subject–verb agreement, tense, and mechanics—matching or slightly exceeding human teachers and traditional grammar checkers (Craven & Fredrick, 2026; Pasha, 2026; Jiang et al., 2023; Lin et al., 2026; Luo et al., 2025).
Moderate	Generative models exhibit a **pronounced overcorrection tendency**, frequently modifying grammatically acceptable phrasings, generating false positives, or altering authentic authorial voice (Jaganov et al., 2025; Wang, 2025; Pasha, 2026; Lin, 2024; Jiang et al., 2023; Zeng et al., 2024).
Moderate	LLMs display **reduced diagnostic reliability** on global discourse coherence, complex rhetorical structure, and non-English target languages, where under-correction and hallucinations occur (Pasha, 2026; Lin, 2024; Pack et al., 2025; Guo, 2024).

FIGURE 2 Evidence strength evaluation for large language model feedback accuracy across error domains

Error Detection Accuracy Across Linguistic Levels

On discrete lexical, grammatical, and syntactic errors, advanced models reliably detect incorrect forms and generate rich metalinguistic explanations (Wang, 2025; Pasha, 2026; Lin et al., 2026). In a benchmark evaluating 560 ungrammatical sentences spanning 20 discrete error types, models including GPT-4, GPT-3.5, Gemini, and Claude 3 significantly outperformed traditional rule-based tools like Grammarly in both identification and correction (Lin et al., 2026). Similarly, in multi-model evaluations of English learner writing, state-of-the-art models reached F1 scores of approximately 0.74, matching trained educator accuracy for local syntactic corrections (Jaganov et al., 2025; Pasha, 2026).

```
flowchart TD
    A[Learner Language Input] --> B{Linguistic Domain}
    B -->|Local Grammar & Syntax| C[High Precision & Recall]
    C --> D[Accurate Metalinguistic Explanations]
    B -->|Global Rhetoric & Coherence| E[Reduced Precision]
    E --> F[Inconsistent & Hallucinated Advice]
    B -->|Advanced Stylistics| G[Risk of Overcorrection]
    G --> H[False Positives & Voice Alteration]
```

Despite high detection rates for isolated sentences, LLM accuracy drops substantially when assessing holistic discourse and argumentation (Craven & Fredrick, 2026; Pasha, 2026). In controlled corpus comparisons, GPT-4 delivered a higher volume of metalinguistic explanations than human instructors but exhibited inconsistent precision, rhetorical redundancy, and occasional hallucinated errors when evaluating extended paragraph logic (Pasha, 2026). Furthermore, comparative trials on college-level English for Academic Purposes essays revealed that GPT-4 was overall less dependable and accurate than experienced human writing instructors, missing nuanced contextual errors and misapplying analytical error codes (Pack et al., 2025).

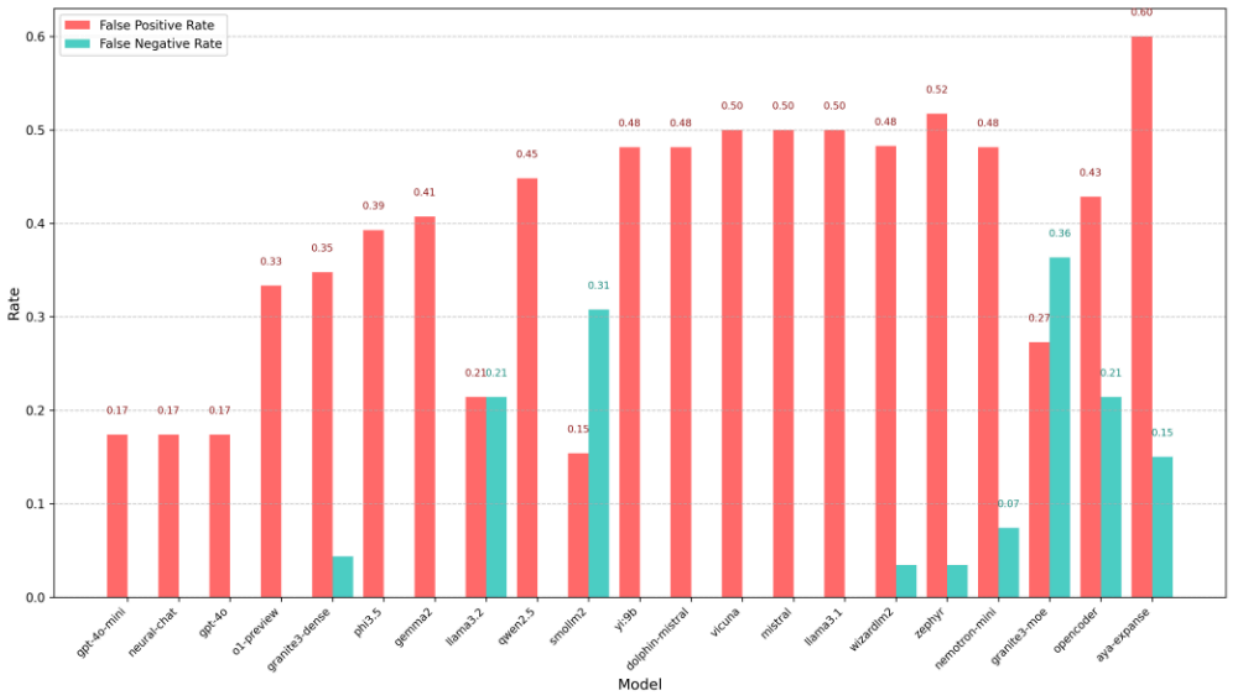


Figure 5. False positive and false negative rates for each LLM, illustrating the models' accuracy in detecting grammatical errors. Models like granite3-moe achieved lower rates, indicating higher precision, while others exhibited higher error rates. (Jaganov et al., 2025)
How to read: The bar chart plots false positive and false negative rates across multiple open-source and proprietary LLMs. Shorter bars indicate superior precision and lower diagnostic error rates.

Overcorrection and Cross-Linguistic Disparities

A persistent limitation of LLMs is their tendency toward overcorrection, leading to high recall but depressed precision (Jaganov et al., 2025; Zeng et al., 2024). Models frequently flag acceptable idiomatic variations or non-standard stylistic choices as errors, which risks confusing developing learners (Jaganov et al., 2025; Jiang et al., 2023; Jaganov et al., 2025). In corpus analyses of Chinese grammatical error correction, LLMs systematically made excessive word-level substitutions rather than minimal phrase edits, resulting in F0.5 scores substantially below dedicated sequence-to-sequence architectures and human annotators (Lin, 2024).

- Overcorrection occurs most frequently in **advanced learner writing**, where models enforce rigid stylistic uniformity onto syntactically valid phrasing (Wang, 2025; Lin, 2024; Zeng et al., 2024).
- Error detection precision in **non-English languages** remains constrained by English-dominant training corpora, causing under-corrections and hallucinations in morphologically complex languages like Chinese and German (Lin, 2024; Guo, 2024).
- Domain-specific prompt engineering and **few-shot examples** dramatically elevate feedback coverage, raising error recall from 10% under generic zero-shot prompts to 55% while maintaining 94% precision (Luo et al., 2025; Zeng et al., 2024).
- Integrating multimodal articulatory representations into speech models enables **precise phoneme diagnosis**, boosting pronunciation feedback accuracy to near state-of-the-art levels (G-Score 0.52 vs. 0.54) (Xie et al., 2026).

Pedagogical Utility and Instructional Processing

The pedagogical value of LLM feedback depends heavily on whether students passively copy automated revisions or actively process the metalinguistic reasoning (Pasha, 2026; Coyne et al., 2025). Second language acquisition research demonstrates that while direct corrective feedback promotes immediate revisions, durable retention requires cognitive engagement, deep processing, or explicit metalinguistic explanations (Coyne et al., 2025; Benson & Dekeyser, 2019; Mujtaba et al., 2021; Suzuki & Saito, 2026; Van Beuningen et al., 2012; Sarré et al., 2019).

```
flowchart LR
    A[LLM Corrective Output] --> B[Direct Rewrite / 1-Click Fix]
    B --> C[Passive Adoption & Shallow Processing]
    C --> D[Ephemeral Revision Gains]
    A --> E[Metalinguistic Hints & Explanations]
    E --> F[Active Cognitive Engagement]
    F --> G[Long-Term Accuracy Development]
```

Because commercial LLM interfaces tend to default to direct, click-to-fix rewrites, they risk acting as cognitive inhibitors that replace the active effort necessary for acquisition (Pasha, 2026; Coyne et al., 2025). Dynamic assessment platforms that constrain LLMs to deliver graduated, progressively explicit hints preserve learner reflection while minimizing false interpretations (Jaganov et al., 2025; Wang, 2025). Consequently, empirical reviews consistently argue for a hybrid model where AI handles rapid, local surface accuracy, while human educators guide discourse structure and safeguard authorial voice (Craven & Fredrick, 2026; Pasha, 2026; Liu & Pagano, 2026; Crosthwaite, 2017).

These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at https://consensus.app. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.

References

Benson, S. D., & Dekeyser, R. (2019). Effects of written corrective feedback and language aptitude on verb tense accuracy. *Language Teaching Research, 23*, 702 - 726. https://doi.org/10.1177/1362168818770921

Coyne, S., Galván-Sosa, D., Spring, R., Guerraoui, C., Zock, M., Sakaguchi, K., & Inui, K. (2025). Annotating Errors in English Learners' Written Language Production: Advancing Automated Written Feedback Systems. *ArXiv, abs/2508.06810*. https://doi.org/10.1007/978-3-031-98459-4_21

Craven, L., & Fredrick, D. R. (2026). LLM-Generated Feedback in L2 Writing: A Scoping Review. *Education Sciences*. https://doi.org/10.3390/educsci16081196

Crosthwaite, P. (2017). Retesting the limits of data-driven learning: feedback and error correction. *Computer Assisted Language Learning, 30*, 447 - 473. https://doi.org/10.1080/09588221.2017.1312462

Guo, X.-S. (2024). Facilitator or thinking inhibitor: understanding the role of ChatGPT-generated written corrective feedback in language learning. *Interactive Learning Environments, 33*, 3526 - 3544. https://doi.org/10.1080/10494820.2024.2445177

Jaganov, T., Blake, J., Villegas, J., & Carr, N. (2025). Large Language Model-Driven Dynamic Assessment of Grammatical Accuracy in English Language Learner Writing. *IEEE Access, 13*, 151538-151550. https://doi.org/10.1109/access.2025.3603191

Jiang, Z.-L., Xu, Z.-X., Pan, Z.-L., He, J.-W., & Xie, K. (2023). Exploring the Role of Artificial Intelligence in Facilitating Assessment of Writing Performance in Second Language Learning. *Languages*. https://doi.org/10.3390/languages8040247

Lin, S. (2024). Evaluating LLMs' grammatical error correction performance in learner Chinese. *PLOS ONE, 19*. https://doi.org/10.1371/journal.pone.0312881

Lin, H. P.-S., Ngo, T. T. N., & Chen, H. H.-J. (2026). Comparative Analysis of LLM-Based Writing Tools for Error Correction and Feedback: A Study on ChatGPT-3.5, ChatGPT-4, Gemini, and Claude 3. *English Teaching & Learning*. https://doi.org/10.1007/s42321-026-00236-4

Liu, W.-J., & Pagano, A. (2026). Leveraging a large language model for error analysis-based automatic feedback in interpreter training. *Translation, Cognition & Behavior*. https://doi.org/10.1075/tcb.00101.liu

Luo, N., Zhang, Z., Zhou, Y., & Zhao, R. (2025). Impact of prompt sophistication on ChatGPT's output for automated written corrective feedback. *ReCALL*. https://doi.org/10.1017/s095834402510044x

Mujtaba, S., Reynolds, B., Parkash, R., & Singh, M. K. A. (2021). Individual and collaborative processing of written corrective feedback affects second language writing accuracy and revision. *Assessing Writing*. https://doi.org/10.1016/j.asw.2021.100566

Pack, A., Hartshorn, K., Escalante, J., & Gillette, N. (2025). How well can GenAI (GPT-4) provide written corrective feedback on English-language learners' writing?. *International Journal of English for Academic Purposes: Research and Practice*. https://doi.org/10.3828/ijeap.2025.2

Pasha, M. (2026). Generative AI as an Automated Written Corrective Feedback Provider in EFL Academic Writing: A Mixed-Methods Investigation of Accuracy Gains, Feedback Quality, and Learner Engagement. *Journal of Artificial Intelligence and Computing*. https://doi.org/10.57041/x2hg0114

Sarré, C., Grosbois, M., & Brudermann, C. (2019). Fostering accuracy in L2 writing: impact of different types of corrective feedback in an experimental blended learning EFL course. *Computer Assisted Language Learning, 34*, 707 - 729. https://doi.org/10.1080/09588221.2019.1635164

Suzuki, W., & Saito, R. (2026). Cognitive and affective reactions to written corrective feedback, type of linguistic items, and second language writing accuracy improvement: an exploratory study. *Language Awareness, 35*, 975 - 995. https://doi.org/10.1080/09658416.2025.2611884

Van Beuningen, C., De Jong, N. D., & Kuiken, F. (2012). Evidence on the Effectiveness of Comprehensive Error Correction in Second Language Writing. *Language Learning, 62*, 1-41. https://doi.org/10.1111/j.1467-9922.2011.00674.x

Wang, J. (2025). EDCEW-LLM: Error detection and correction in English writing: A large language model-based approach. *Alexandria Engineering Journal*. https://doi.org/10.1016/j.aej.2025.08.005

Xie, Y., Zhong, H., Lan, X.-H., & Dong, W. (2026). Mispronunciation detection and diagnosis based on large language models. *Comput. Speech Lang., 99*, 101942. https://doi.org/10.1016/j.csl.2026.101942

Zeng, M., Kuang, J., Qiu, M., Song, J., & Park, J. (2024). Evaluating Prompting Strategies for Grammatical Error Correction Based on Language Proficiency. 6426-6430. https://doi.org/10.48550/arxiv.2402.15930